

# Model Performance Test

## Artificial Intelligence

Project Name: AI-Enhanced Intrusion Detection System

Made by

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## Model Performance Testing

Project team has filled the following information in the model performance testing template.

| S.No. | Parameter                             | Values                                                                                                                                                                                                      |
|-------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | Model Summary                         | Random Forest Classifier trained on CICIDS2017 Dataset. Optimized to 4 key features (Flow Duration, Total Fwd Packets, Total Bwd Packets, Total Length of Fwd Packets). Class imbalance handled with SMOTE. |
| 2.    | Accuracy                              | Training Accuracy – ~99.2% Validation Accuracy – ~98.9%                                                                                                                                                     |
| 3.    | Confidence Score (Only YOLO Projects) | Class Detected – N/A (Non-YOLO project) Confidence Score – N/A                                                                                                                                              |

## Notes

- ▶ The model uses Random Forest with 4 features: Flow Duration, Total Fwd Packets, Total Bwd Packets, Total Length of Fwd Packets.
- ▶ SMOTE was applied to handle class imbalances in the CICIDS2017 dataset.
- ▶ Verdict classes: BENIGN, Web Attack – Brute Force, Web Attack – SQL Injection, Web Attack – XSS.

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